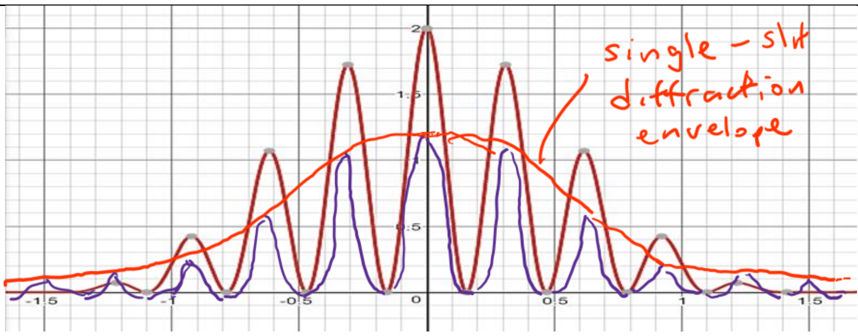
Q9		
(a)	When two or more waves overlap/meet at the point (at a particular instance), the **resultant displacement** at that point is the **vector sum of the displacements** which would be caused by each wave at the point (at that instance).	**B1** **B1**
(b)(i)	From graph, Period of waves = 40.0×10^{-9} s Frequency = $1/40.0 \times 10^{-9} = 25.0 \times 10^{6}$ Hz = 25.0 MHz	 **B1** **A0**
(b)(ii)	At M, the waves arrive in phase / path difference is zero, hence constructive interference occurs Resultant wave amplitude = 2A (where A is the amplitude due to an individual wave), A = 0.5 units When only one source is on, the amplitude is 0.5 units. Diagram of waveform with same period, phase and amplitude = 0.5 units drawn. B1 – explain why constructive interference B1 – either indicating that resultant wave amplitude is twice or that the amplitude of each wave (from A or B) is half that observed at M. B1 – correct graph drawn	**B1** **B1** **B1**
(b)(iii)	Distance from M to N = distance between an antinode to a node $= \frac{1}{4} \lambda$ $= \frac{1}{4} (c/f)$ $= \frac{1}{4} (3.00 \times 10^{8} / 25.0 \times 10^{6})$ = 3.00 m M1 – relating MN to $\lambda/4$ M1 – for finding λ using $c=f\lambda$ Alternative: Let the distance moved be *x* so that the path difference increased by half a wavelength. $(AN - BN) = \frac{1}{2} \lambda$ $(6.00 + x) - (6.00 - x) = (0.5)(3.00 \times 10^{8} / 25.0 \times 10^{6})$ where *x* is distance from M to N. $x = 3.00$ m	 **M1** **M1** **A0**

(b)(iv)	Point M is 6.00 m from A and B respectively. Point N is 9.00 m from A and 3.00 m from B As A and B are point sources, intensity of wave from each source, $I = \frac{\text{Power from source}}{4\pi r^2}$ (As power from sources are equal and constant), Hence $I \propto \frac{1}{r^2}$ $\frac{I_A}{I} = (\frac{6.00}{9.00})^2$ $I_A = 0.4444I$ $= 0.444$ I (shown) $\frac{I_B}{I} = (\frac{6.00}{3.00})^2$ I_B = 4.00 I (shown)	**B1** **B1** **B1**
(b)(v)	Using $I \propto A^2$ At point N, A_A is amplitude of wave due to source A and A_B is amplitude of wave due to source B. At point M, I is the intensity of wave from a single source (either A or B) and the amplitude of a wave from either source is 0.5 units. $\frac{0.444I}{I} = (\frac{A_A}{0.5})^2$ A_A = 0.333 units $\frac{4I}{I} = (\frac{A_B}{0.5})^2$ A_B = 1.00 units At N, waves source A and B arrive in antiphase, resultant amplitude = 1.000 – 0.333 = 0.666 units.	**M1** **M1** **A1**
(c)(i)	1. *D* must be much larger than *a*. (so that the two paths are parallel, resulting in $a\sin\theta = n\lambda$, where θ is the angle of each order. Clearly from the equation, the orders are not equally spaced). 2. *a* must be much larger than λ. (so that the angle is small and small-angle approximations can be made and fringe separation is then constant).	**A1** **A1**
(c)(ii)	$x = \frac{\lambda D}{a}$ (students must use the symbols defined in the question)	**A1**
(c)(iii)	As the slits have a finite width, the 1st order minima (due to single-slit diffraction) coincides where the 5th order maxima (due to double-slit interference) occurs.	**A1**

(c)(iv)



Marking points:

- All **fringes** should be clearly shown to **occur at the same position as original** diagram. (As fringe separation x does not change.) **(This mark must be awarded for the next mark to be awarded)** **A1**
- **5th order maxima** should be **visible** **A1**

  (Using $\sin\theta = \dfrac{\lambda}{b}$ (single-slit diffraction), when b decreases, the angle of diffraction (of 1st or minima) gets wider thus the 5th order maxima of the double-slit interference pattern should be visible.

Additional points (not assessed)

- The fringes should not exceed the single-slit diffraction envelope. (Also there should be symmetry of intensity about the zero order maximum.
- As *b* decreases, amount of energy that passes through slits decreases, thus the intensity of light at the zero order maximum (for the double-slit interference pattern) should be reduced. For example, if the width is halved, the amplitude at the central maximum would be halved, resulting in intensity that is ¼ of the original.

Q10		